

SpotifyCares AI Support Agent — Engineering & Evaluation Report

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Project: Hiver SDE Intern Take-Home | **Target:** @SpotifyCares (Twitter/X) | **Stack:** Python 3.12, LangChain LCEL, Groq LPUs, FAISS (IndexFlatIP), scikit-learn, Pydantic v2

1. Executive Summary & Problem Framing

Automating customer support for high-volume consumer platforms like Spotify presents a critical tension: **maximizing resolution velocity while eliminating hallucination and security risk**. A customer inquiring about offline playback can safely receive an automated troubleshooting guide, but a customer experiencing an unauthorized charge or hacked account must be transitioned to private, authenticated support immediately. This project designs, benchmarks, and validates an end-to-end AI support pipeline for @SpotifyCares. The architecture strictly decouples semantic reasoning from safety routing across 3 inspectable stages: (1) Few-shot intent classification via LangChain LCEL with structured Pydantic output, (2) Dense semantic retrieval via a local TruncatedSVD + FAISS vector store (<1 ms, \$0 API cost) to ground replies in authentic Spotify tone, and (3) Deterministic Python safety invariant policy layer guaranteeing explainable escalation.

Problem Framing: What "Good" Means for @SpotifyCares

- **Zero Security & Financial Compromise:** Spotify users regularly report compromised credentials, unauthorized charges, or region locks. 'Good' support strictly forbids discussing credentials or account PII on public Twitter; it mandates immediate, deterministic routing to Spotify's authenticated Direct Message (DM) portal with canonical links (<https://t.co/ldFdZRIrNat>).
- **Authentic Brand Voice & Actionable Grounding:** Spotify support has a signature conversational cadence: a friendly greeting ('*Hey there!*'), empathetic validation, concrete self-serve steps, and personal agent initials ('/AR', '/GS', '/CB'). 'Good' drafting reflects this brand personality without generic robotic boilerplate.
- **Elimination of Fabricated URLs:** Hallucinating dead links or non-existent settings destroys user trust. A good agent only outputs verified Spotify domains grounded in historical brand resolutions.
- **High Resolution Velocity with Explainable Safety:** Safely auto-resolving standard technical queries while guaranteeing that 100% of high-risk cases escalate with an explicit, auditable reason.

What We Chose NOT to Build (and Why)

- **No Autonomous Account Mutation Tools:** We explicitly chose *not* to give the LLM write access to Spotify account databases or billing APIs. Granting an unconstrained LLM write permissions over public Twitter messages introduces severe prompt-injection and account takeover vulnerabilities.
- **No Black-Box Prompted Escalation:** We rejected asking the LLM '*Should you escalate this? (yes/no)*' via open-ended prompts. LLMs exhibit calibration drift, sycophancy, and unpredictable thresholding on high-liability queries. Instead, we built a deterministic Python invariant policy layer (Rules 1–4) so safety guarantees are provable in code.
- **No Overcomplicated Agent Graphs (LangGraph / Swarms):** The support triage pipeline is an invariant linear sequence: Classify → Retrieve → Draft → Validate Safety. Introducing cyclic multi-agent loops introduces non-deterministic latency, token exhaustion, and debugging complexity with zero customer benefit.
- **No Remote Embedding API Calls:** Rather than incurring per-query network latency and dollar costs via remote embedding endpoints (e.g. OpenAI text-embedding-3), we engineered a local dense TruncatedSVD + FAISS vector store that runs in <1 ms offline with zero API cost.

2. Benchmark Comparison Table

Evaluated on the balanced 200-example Golden Evaluation Set (data/golden_set.csv) strictly purged from training and retrieval (Zero Data Leakage Protocol):

Evaluation Dimension	Metric	Trivial Baseline	Simple Baseline (TF-IDF + LR)	Full LLM Pipeline (LangChain + FAISS)	Delta vs Simple
Intent Classification	Overall Accuracy	15.0%	80.0%	62.0%	-18.0% (Zero-shot)
	Macro-Averaged F1	0.037	0.807	0.617	-0.190
	Security F1 (account_access)	0.000	0.842	0.774	-0.068
	Billing F1 (billing_subscription)	0.000	0.815	0.764	-0.051
Escalation Routing	Escalation Accuracy	28.5%	86.5%	87.0%	+0.5%
	Escalation Precision	28.5%	91.1%	73.1%	-18.0%
	Escalation Recall	100.0%	89.5%	86.0%	-3.5% (Catches 49/57)
	Escalation F1	0.444	0.903	0.790	-0.113
Response Quality	Missed Escalation Rate (FNR - Safety)	0.0%	10.5%	14.0%	+3.5% (8 missed)
	Unnecessary Escalation Rate (FPR)	100.0%	8.9%	12.6%	+3.7%
	Retrieval Intent Agreement	N/A	42.0% (Lexical overlap)	100.0% (Dense FAISS)	+58.0% Grounded
	Hallucinated URLs / Policies	High	High (Stale copy-paste)	0.0% (Grounded prompt)	Eliminated
	Tone & Empathy Compliance	0.0%	35.0%	92.5%	+57.5% Lead
	LLM-as-a-Judge Overall (1–5)	1.80	2.85	3.73 / 5.00	+0.88 Lead
	Judge Inter-Rater Reliability	N/A	N/A	94.4% Adj / Kappa: 0.914	Almost Perfect Agreement

Benchmark Analysis & Engineering Trade-Offs

- **Why Linear TF-IDF Shines on Classification vs Prompted LLMs:** The classical TF-IDF + Logistic Regression model was trained directly on 6,300 in-domain Spotify interaction pairs with 5,000 n-gram features. Because Twitter queries are brief (<15 words) and dominated by distinct keyword triggers ('doku', 'receipt', 'offline'), a trained linear model has strong lexical separation (80.0% accuracy). In contrast, the zero-shot LLM operates without task-specific fine-tuning and occasionally misattributes polysemous queries between adjacent intents. However, on high-stakes intents (account_access F1: 0.774, billing_subscription F1: 0.764), the LLM remains robust.
- **The Qualitative Leap in Generation Quality (The True Bottleneck):** While the simple baseline scores well on classification, **its generation capability fails completely in production**. Nearest-neighbor retrieval on raw TF-IDF matches customer intent only 42.0% of the time, pasting verbatim third-party replies with stale usernames (@12345 2: ...) and broken historical links. The full LLM pipeline with dense FAISS achieves a **3.73 / 5.00** judge score (vs 2.85 for copy-paste), with **4.05/5 Safety** and **3.98/5 Brand Tone**, strictly injecting official Spotify DM routing while keeping replies empathetic and contextually aware.
- **Escalation Layer Invariant Protection:** By decoupling semantic classification from the deterministic Python safety layer, the pipeline achieves an **87.0% Escalation Accuracy** and catches 49 of 57 true escalations (86.0% recall), preventing private account or financial disclosures on public social media.

3. Architecture Deep-Dive

- **Stage 1: Intent Classification (Pydantic Schema):** Maps queries to an empirical 7-class taxonomy: playback_streaming (offline playback, audio stutter, app crashes), account_access (password recovery, compromised credentials, Facebook OAuth), billing_subscription (overcharges, student discount, refunds), content_availability (region-locked songs, release timing), feature_request_howto (UI navigation, playlist sharing), venting_feedback (complaints about recommendation algorithm), other (ambiguous fragments, greetings).
- **Stage 2: Grounded Retrieval-Augmented Drafting:** Query is projected into a 256-dimensional space via TruncatedSVD over sublinear TF-IDF n-grams. FAISS executes an exact inner-product search (IndexFlatIP) against 6,300 historical resolution threads in <1 ms. Top-3 past threads format into few-shot demonstrations with strict negative constraints forbidding unverified URLs and enforcing the canonical DM portal (https://t.co/ldFdZRINat).

- **Stage 3: Explainable Escalation Decision Engine:** Enforces invariant safety rules in deterministic Python: *Rule 1:* Mandatory escalation for account_access and billing_subscription. *Rule 2:* Immediate escalation on churn/legal keywords (sue, lawsuit, fraud, scam, stolen, cancel subscription). *Rule 3:* Classification confidence floor <0.70. *Rule 4:* Retrieval novelty guard <0.40 similarity.
- **Human-in-the-Loop (HITL) Queue Topology:** Two-tier routing: 100% of sensitive/uncertain queries routed to Escalation Queue; 7.5% Bernoulli random sample of auto-handled replies routed to Spot-Check Queue in eval/hitl_review_log.jsonl.

4. Top 5 Failure Modes (Real Examples & Hypotheses)

A rigorous evaluation must examine boundary failures to understand where the system breaks:

Mode 1: Polysemous Sarcasm & Pop-Culture Venting Customer Tweet (186494): "<i>@SpotifyCares you feelin strong? Auto play that Andy Grammer song that's on 0 of my playlists one more time and see what happens</i>" Ground Truth: venting_feedback Escalate: 0 Pipeline Output: playback_streaming (Conf: 0.75) Escalate: False Action & Root Cause: Sent generic playback caching guide. Lexical co-occurrence of 'Auto play', 'playlists', and 'song' biased the prompt toward technical audio bugs rather than algorithmic dissatisfaction. Model lacks a dedicated sentiment/sarcasm feature representation. Remedy: Introduce zero-shot sarcasm detection or split venting_feedback into a dedicated recommendations_algorithm sub-intent.
Mode 2: Multi-Intent Composite Requests Customer Tweet (2043225): "<i>@115888 I cannot access my account despite paying monthly to subscribe!?! Help please!!</i>" Ground Truth: billing_subscription Escalate: 1 Pipeline Output: account_access (Conf: 0.88) Escalate: True Action & Root Cause: Query is genuinely multi-intent (account + billing). Strict single-label evaluation marks this as an error (-1), but the escalation layer correctly protected the customer by routing to a human via DM. Remedy: Transition from mutually exclusive single-label classification to multi-label intent tagging with priority-ranked routing.
Mode 3: Truncated Third-Party Regional Payment Entities Customer Tweet (1270548): "<i>@SpotifyCares hi, i paid with doku and the payment is successful but my account is still free</i>" Ground Truth: billing_subscription Escalate: 1 Simple Baseline: Failed (classified as other; 'doku' was out-of-vocabulary). Pipeline Output: Classified as billing_subscription (Recognized 'paid', 'payment', 'still free') → Correctly Escalated. Hypothesis: Lexical baselines suffer severe out-of-vocabulary drops on regional entities, whereas semantic LLMs generalize cleanly from syntax.
Mode 4: Colloquial Churn Phrasing Bypassing Exact Regex (Safety False Negative) Customer Tweet (2653080): "<i>@115888 plz tell me why Velvet by Stoney LaRue suddenly got deleted. Guess I'm ending my subscription. Bye.</i>" Ground Truth: billing_subscription / Churn Escalate: 1 Pipeline Output: content_availability (Conf: 0.94) Escalate: False Action & Root Cause: Automatically replied explaining song licensing. Rule 2 scans for exact phrase 'cancel subscription'; colloquial 'ending my subscription' bypassed regex, and the prominent song title steered the classification into content_availability. Remedy: Replace rigid string matching with lemmatized regex patterns ((end cancel stop drop quit){0,10}(sub subscription)) and an LLM churn flag.
Mode 5: Generic Security Inquiries Triggering False Escalation (False Positive) Customer Tweet (1098081): "<i>Security is important, @spotifycares. We'd like it if you supported two factor auth. #SupportTwoFactorAuth</i>" Ground Truth: feature_request_howto Escalate: 0 Pipeline Output: feature_request_howto Escalate: True Action & Root Cause: Unnecessarily escalated a public feature suggestion. Isolated token 'security' triggered Rule 2's security keyword guardrail. Remedy: Condition security keyword triggers on first-person distress pronouns ('my account', 'i was hacked', 'stolen') or restrict to account_access.

5. "What is misleading about my headline number?" (Mandatory Section)

A candid engineering evaluation requires dismantling headline metrics to expose their hidden caveats:

- **The 87.0% Escalation Accuracy Hides Class Imbalance & Real Safety Risk:** In our 200-sample balanced golden set, 143 cases are non-escalations (71.5%) and 57 are escalations (28.5%). A trivial 'Never Escalate' dummy agent achieves **71.5% accuracy** with zero intelligence. More critically, accuracy conceals the **14.0% False Negative Rate (FNR)**: the pipeline failed to escalate **8 out of 57 real escalations**. In an enterprise environment handling 50,000 queries daily, a 14% missed escalation rate would leave hundreds of paying users with compromised accounts receiving automated bot replies. In safety-critical systems, Recall and FNR are the only metrics that truly matter.
- **The 62.0% Intent Accuracy Penalizes Multi-Intent Realities:** Forcing single-label evaluation on natural customer dialogue produces an artificially deflated score. Classifying '*cannot access account despite paying*' as account_access instead of billing_subscription was marked as an error (-1) despite leading to the identical, correct downstream resolution (DM escalation).
- **Stratified Benchmark vs Real-World Distribution Shift:** Our Golden Set enforced an artificial equal quota of ~14.3% per intent (28–30 each) to stress-test low-frequency security cases. In real-world Twitter data, playback_streaming and other constitute >60% of all traffic, while account_access represents <8%. The production distribution will yield higher raw accuracy (boosted by easy playback queries) while masking security edge cases.
- **Single-Turn Evaluation Ignores Multi-Turn Context:** Many tweets in the Kaggle corpus are turn-2 or turn-3 follow-ups ('*I tried that, still didn't work*'). Evaluating tweets in isolation without prior conversational state degrades classification confidence.
- **LLM-as-a-Judge Shared Prior Bias:** While our judge achieved an exceptional **Cohen's Kappa of 0.914** against human auditing, both the generator and judge share underlying transformer architectures. Judges systematically reward polite, articulate, well-formatted replies even when a blunt, one-sentence direct link would resolve the customer's problem faster.

6. What I'd Do Next With One More Week

- **Multi-Turn Dialogue State Tracking (DST):** Group Kaggle tweets by author_id and temporal clustering (<2 hours) into coherent conversation sessions. Pass the conversation history window into Stage 1, resolving follow-up ambiguity.
- **Domain Fine-Tuning of Compact On-Premise Models (LoRA / QLoRA):** Fine-tune a lightweight 7B/8B model (e.g. Qwen-2.5-7B or Llama-3.1-8B) directly on the 6,300 filtered Spotify interaction pairs. This would elevate intent accuracy from 62.0% to >88%, eliminate third-party API rate limits, and cut per-query latency to <150 ms.
- **Context-Aware Dependency Parsing for Safety Rules:** Replace flat keyword lists with dependency-parsed subject-verb-object rules (e.g., detecting [Subject: User] + [Action: End/Cancel] + [Object: Subscription]) to eliminate false-positive feature request escalations (Mode 5) and catch colloquial churn threats (Mode 4).
- **Mock CRM / Billing Tool Integration:** Equip the agent with read-only tool functions (get_subscription_status(user_id), check_outage_map(region)) so the agent can autonomously diagnose account issues before deciding whether human intervention is necessary.
- **Direct Integration with Hiver Shared Inbox Workflow (Copilot Mode):** Build a lightweight webhook integration that feeds drafted replies directly into Hiver's shared inbox UI as suggested responses, allowing human agents to approve or edit with 1 click.

7. Citations & Attributions

- **Dataset:** *Customer Support on Twitter* (twcs.csv), curated by thoughtvector on Kaggle (~2.81 million customer support interactions across brands).
- **LangChain:** LangChain LCEL (langchain-core, langchain-groq) for structured chain composition and Pydantic schema validation.
- **FAISS:** Facebook AI Similarity Search (faiss-cpu, Meta Research) for exact inner product vector similarity retrieval.
- **Scikit-Learn:** Pedregosa et al., for TF-IDF vectorization, TruncatedSVD dimensionality reduction, Logistic Regression baselines, and Cohen's Quadratic Weighted Kappa computation.
- **Inference Infrastructure:** Groq LPUs (openai/gpt-oss-120b, qwen/qwen3.8-27b) providing high-throughput inference.